

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Uses partial fractions with linear denominators $\frac{6x+1}{6x^2-7x+2} = \frac{A}{ax+b} + \frac{B}{cx+d}$	AO3.1a	M1	$\frac{6x+1}{6x^2-7x+2} = \frac{A}{3x+2} + \frac{B}{2x-1}$ $A(2x-1) + B(3x-2) = 6x+1$
Obtains correct linear denominators	AO1.1b	B1	$x = \frac{2}{3}, A\left(\frac{1}{3}\right) = 5 \text{ so } A = 15$
Obtains at least one numerator correct (using any valid method, eg equating coefficients or substitution of values)	AO1.1b	A1	$x = \frac{1}{2}, B\left(-\frac{1}{2}\right) = 4 \text{ so } B = -8$ $\int_1^2 \frac{15}{3x-2} - \frac{8}{2x-1} dx$ $= [5\ln(3x-2) - 4\ln(2x-1)]_1^2$
Obtains partial fractions completely correct	AO1.1b	A1	$= 5\ln(4) - 4\ln(3) - (5\ln(1) - 4\ln(1))$ $= 10\ln(2) - 4\ln(3)$
Integrates 'their' partial fractions, must include logs $p\ln(ax+b) + q\ln(cx+d)$	AO1.1a	M1	
'Their' integral correct (ignore limits)	AO1.1b	A1F	
Substitutes limits into 'their' integral	AO1.1a	M1	
Correct final answer in correct form CAO	AO1.1b	A1	
Total 8 marks			

Q2.

(a) $7x - 1 = A(1 + 3x) + B(3 - x)$

M1

$$x = 3 \quad x = -\frac{1}{3}$$

Use two values of x to find A and B .

Or solve $A + 3B = -1 \quad 3A - B = 7$

Or cover up rule

m1

$$A = 2 \quad B = -1$$

A1

3

(b) (i) $\frac{1}{1+3x} = (1+3x)^{-1} = 1 + (-1)3x + \frac{1}{2}(-1)(-2)(3x)^2$
Condone missing brackets

M1

$$= 1 - 3x + 9x^2$$

A1

$$\frac{1}{3-x} = (3-x)^{-1} = \frac{1}{3} \left(1 - \frac{x}{3} \right)^{-1}$$

B1

$$\left(1 - \frac{x}{3} \right)^{-1} = 1 + (-1) \left(-\frac{x}{3} \right) + kx^2$$

Condone missing brackets

M1

$$= 1 + \frac{x}{3} + \frac{x^2}{9}$$

A1

$$\frac{7x-1}{3+8x-3x^2} = 2 \times \frac{1}{3} \times \left(1 + \frac{x}{3} + \frac{x^2}{9} \right) - 1 \times (1-3x+9x^2)$$

*Attempt to use PFs to combine expansions, or expand
 $(7x-1)(3-x)^{-1}(1+3x)^{-1}$
 and simplify to $a+bx+cx^2$*

M1

$$= -\frac{1}{3} + \frac{29}{9}x - \frac{241}{27}x^2$$

A1

7

(ii) 0.4 is outside the range of validity, because $0.4 > \frac{1}{3}$.

$$\text{OE Accept } 0.4 > \frac{1}{3}$$

B1

1

[11]

Q3.

(a) (i) $5 - 8x = A(1 - 3x) + B(2 + x)$

M1

$$x = -2 \quad x = \frac{1}{3}$$

Two values of x used to find values for A and B

m1

$$A = 3 \quad B = 1$$

A1

3

Alternative:

$$5 - 8x = A(1 - 3x) + B(2 + x)$$

(M1)

$$5 = A + 2B$$

$$-8 = -3A + B$$

Set up simultaneous equations and solve.

(m1)

$$A = 3 \quad B = 1$$

(A1)

(3)

(ii) $\int_{-1}^0 \frac{3}{2+x} + \frac{1}{1-3x} \, dx = 3 \ln(2+x) - \frac{1}{3} \ln(1-3x)$

$a \ln(2+x) + b \ln(1-3x)$ where a and b are constants

M1

$$= (3 \ln 2 - \frac{1}{3} \ln 1) - (3 \ln 1 - \frac{1}{3} \ln 4)$$

$f(0) - f(-1)$ used

m1

$$= 3 \ln 2 + \frac{1}{3} \ln 4$$

ft A and B

A1ft

$$= \frac{11}{3} \ln 2$$

ft $\left(A + \frac{2}{3}B\right) \ln 2$

A1ft

4

(b) (i) $(C =)2$

B1

1

(ii) $\int \frac{9-18x-6x^2}{2-5x-3x^2} dx = \int C dx + \int \frac{5-8x}{2-5x-3x^2} dx$

Seen or implied.

Allow $\pm C + \int \frac{5-8x}{2-5x-3x^2} dx$

M1

$$\int_{-1}^0 \frac{9-18x-6x^2}{2-5x-3x^2} dx = 2 + \frac{11}{3} \ln 2$$

Accept $2 + 3 \ln 2 + \frac{1}{3} \ln 4$

ft 2 + candidate's answer to part (a)(ii) if exact.

A1ft

2

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

[10]

Q4.

(a) $2x + 3 = A(2x + 1) + B(2x - 1)$

M1

$$x = \frac{1}{2} \quad x = -\frac{1}{2}$$

Use two values of x to find A and B

m1

$$A = 2 \quad B = -1$$

Both

A1

3

Alternative; equating coefficients

$$2x + 3 = A(2x + 1) + B(2x - 1)$$

(M1)

$$x \text{ term} \quad 2 = 2A + 2B$$

$$\text{constant} \quad 3 = A - B$$

Set up simultaneous equations and solve.

(m1)

$$A = 2 \quad B = -1$$

Both

(A1)

(3)

Alternative; cover up rule

$$x = \frac{1}{2} \quad A = \frac{2 \times \frac{1}{2} + 3}{2 \times \frac{1}{2} + 1} \quad \left(= \frac{4}{2} \right)$$

$$x = \frac{1}{2} \quad \text{and} \quad x = -\frac{1}{2} \quad \text{used to find } A \text{ and } B$$

(M1)

$$x = -\frac{1}{2} \quad B = \frac{2 \times (-\frac{1}{2}) + 3}{2 \times (-\frac{1}{2}) - 1} \quad \left(= \frac{2}{-2} \right)$$

$$A = 2 \quad B = -1$$

A and B both correct 3 / 3

One of A or B correct 1 / 3

(A1A1)

(3)

Condone poor algebra for M1 if continues correctly.

(b)

$$\begin{array}{r}
 3x \\
 4x^2 - 1 \overline{) 12x^3 - 7x - 6} \\
 \underline{12x^3 - 3x} \\
 - 4x - 6
 \end{array}$$

Complete division leading to values for C and D

M1

$$C = 3$$

A1

$$D = -2$$

A1

$C = 3, D = -2$ stated or written in expression.

SC B1

$C = 3, D$ not found or wrong;

$D = -2, C$ not found or wrong.

3

Alternative

$$\frac{12x^3 - 7x - 6}{4x^2 - 1} = \frac{12x^3 - 3x - 4x - 6}{4x^2 - 1} = 3x - \frac{2(2x+3)}{4x^2 - 1}$$

(M1)

$$C = 3$$

(A1)

$$D = -2$$

(A1)

$C = 3, D = -2$ stated or written in expression.

SC B1

$C = 3, D$ not found or wrong;

$D = -2, C$ not found or wrong.

(3)

Alternative

$$12x^3 - 7x - 6 = 4Cx^3 - Cx + 2Dx + 3D$$

Complete method for C and D

(M1)

$$C = 3$$

(A1)

$$D = -2$$

(A1)

$C = 3, D = -2$ stated or written in expression.

SC B1

$C = 3, D$ not found or wrong;

$D = -2, C$ not found or wrong.

(3)

Alternative

$$x = 0 \quad x = 1$$

$$6 = -3D - \frac{1}{3} = C + \frac{5}{3}D$$

Use two values of x to set up simultaneous equations

(M1)

$$C = 3$$

(A1)

$$D = -2$$

(A1)

$C = 3, D = -2$ stated or written in expression.

SC B1

$C = 3, D$ not found or wrong;

$D = -2, C$ not found or wrong.

(3)

Complete division for M1; obtain a value for $C(Cx)$ and a remainder $ax + b$

(c) $\int 3x - 2 \left(\frac{2}{2x-1} - \frac{1}{2x+1} \right) dx$

Use parts (a) and (b) to obtain integrable form

M1

$$3 \frac{x^2}{2}$$

ft on C

A1ft

$$-2 \left(\ln(2x-1) - \frac{1}{2} \ln(2x+1) \right)$$

Both correct; ft on A, B and D

Condone missing brackets

A1ft

$$\frac{3}{2}(4-1) - 2 \left(\left(\ln 3 - \frac{1}{2} \ln 5 \right) - \left(\ln 1 - \frac{1}{2} \ln 3 \right) \right)$$

Correct substitution of limits

m1

$$\frac{9}{2} - 3\ln 3 + \ln 5 = \frac{9}{2} + \ln\left(\frac{5}{27}\right)$$

$$p = \frac{9}{2} \quad q = \frac{5}{27}$$

A1

Form $\int Cx + \left(\frac{P}{2x-1} + \frac{Q}{2x+1} \right) dx$ using candidate's P, Q, C for M1.

Condone missing dx .

$$\int Cx \, dx = C \frac{x^2}{2} \text{ for A1ft}$$

ISW extra terms eg $\frac{12}{4x^2-1}$ for first three terms only; max 3/5.

Candidate's C ; must have a value.

$$\int \frac{4x+6}{4x^2-1} \, dx = \int \frac{4x}{4x^2-1} + \frac{6}{4x^2-1} \, dx \text{ is an integrable form,}$$

as $\int \frac{1}{x^2-a^2} \, dx = \frac{1}{2a} \ln\left(\frac{x-a}{x+a}\right)$ is in the formula book, but they **must** try to

integrate to show they know this, **or** use partial fractions again with

$$\frac{6}{4x^2-1} = \frac{3}{2x-1} - \frac{3}{2x+1} \text{ for M1.}$$

Substitute limits into $C \frac{x^2}{2} + m\ln(2x-1) + n\ln(2x+1)$, or equivalent, for m1;
substitution must be completely correct.

$$\text{Condone } \frac{9}{2} - \ln\left(\frac{27}{5}\right) \text{ for A1.}$$

5

[11]

Q5.

(a) (i) $5x - 6 = A(x - 3) + Bx$

Multiply by denominator and use two values of x .

M1

$$x = 0 \quad x = 3$$

$$A = 2 \quad B = 3$$

A1

Alternative: equate coefficients

$$-6 = -3A \quad 5 = A + B$$

Set up and solve simultaneous equations for values of A and B.

(M1)

$$A = 2 \quad B = 3$$

(A1)

2

(ii) $\left(\int \frac{2}{x} + \frac{3}{x-3} dx = \right) 2 \ln x$
their A ln x

B1ft

$$+ 3 \ln(x-3) (+C)$$

their B ln (x - 3) and no other terms; condone B ln x - 3

B1ft

2

(b) (i)

$$\begin{array}{r}
 2x+1 \overline{) 4x^3 + 5x - 2} \\
 \underline{4x^3 + 2x^2} \\
 -2x^2 + 5x \\
 \underline{-2x^2 - x} \\
 6x - 2 \\
 \underline{6x + 3} \\
 -5
 \end{array}$$

Division as far as $2x^2 + px + q$ with $p \neq 0$, $q \neq 0$, PI

M1

$$p = -1$$

PI by $2x^2 - x + q$ seen

A1

$$q = 3$$

PI by $2x^2 - x + 3$ seen

A1

$$r = -5$$

and must state $p = -1, q = 3, r = -5$ explicitly or write out full correct RHS expression

A1

Alternative 1:

$$4x^3 + 5x - 2 = 4x^3 + (2 + 2p)x^2 + (p + 2q)x + q + r$$

$$2 + 2p = 0$$

Clear attempt to equate coefficients, PI by $p = -1$

(M1)

$$p + 2q = 5$$

$$q + r = -2$$

$$p = -1$$

(A1)

$$q = 3 \quad r = -5$$

(A1A1)

Alternative 2:

$$4x^3 + 5x - 2 = (2x + 1)(2x^2 + px + q) + r$$

$$x = -\frac{1}{2} \quad 4 \times \left(-\frac{1}{2}\right)^3 + 5 \left(-\frac{1}{2}\right) + 2 = r$$

$$x = -\frac{1}{2} \text{ used to find a value for } r$$

(M1)

$$r = -5$$

(A1)

$$p = -1, q = 3$$

(A1A1)

4

$$(ii) \quad \left(\frac{4x^3 + 5x - 2}{2x + 1} = \right) 2x^2 + px + q + \frac{r}{2x + 1}$$

M1

$$\frac{2}{3}x^3 - \frac{1}{2}x^2 + 3x + k \ln(2x + 1) (+ C)$$

ft on p and q

A1ft

$$\frac{2}{3}x^3 - \frac{1}{2}x^2 + 3x - \frac{5}{2}\ln(2x+1) + C$$

CSO

A1

3

[11]

Q6.

(a) $3 + 9x = A(3 + 5x) + B(1 + x)$

PI by correct A and B

M1

$$x = -1 \quad -\frac{3}{5}$$

Substitute two values of x and solve for A and B.

m1

$$A = 3 \quad B = -6$$

A1

3

Alternative

Equating coefficients

$$3 + 9x = A(3 + 5x) + B(1 + x)$$

©2025 Exam Papers Practice. All rights reserved.

(M1)

$$3 = 3A + B$$

$$9 = 5A + B$$

Set up simultaneous equations and solve.

Condone 1 error.

(m1)

$$A = 3 \quad B = -6$$

(A1)

(3)

Alternative

Cover up rule

$$x = -1 \quad A = \frac{3-9}{3-5}$$

$$x = -\frac{3}{5} \quad B = \frac{3 - \frac{27}{5}}{1 - \frac{3}{5}}$$

$x = -1$ and $x = -\frac{3}{5}$ and attempt to find A and B .

(M1)

$$A = 3 \quad B = -6$$

SC NMS

A and B both correct; 3/3

One of A and B correct 1/3

(A1 A1)

(3)

(b) $(1+x)^{-1} = 1 - x + kx^2$

$$= 1 - x + x^2$$

M1

$$(3+5x)^{-1} = 3^{-1} \left(1 + \frac{5}{3}x\right)^{-1}$$

A1

$$\left(1 + \frac{5}{3}x\right)^{-1} = 1 - \frac{5}{3}x + \left(\frac{5}{3}x\right)^2$$

B1

$$\frac{3}{5} = 1 - \frac{5}{3}x + \frac{25}{9}x^2$$

Condone missing brackets; allow one sign error

M1

$$\frac{3+9x}{(1+x)(3+5x)}$$

A1

$$= 3(1 - x + x^2) - 6 \times 3^{-1} \left(1 - \frac{5}{3}x + \frac{25}{9}x^2 \right)$$

Use PFs and simplify to $a + bx + cx^2$
 or expand product of $(3 + 9x)$ and binomial
 expansions and simplify to $a + bx + cx^2$

M1

$$= 1 + \frac{1}{3}x - \frac{23}{9}x^2$$

A1

7

(c) $\frac{5x}{3} < 1$ oe or $\frac{5x}{3} > -1$ oe
 Condone $\leq$ instead of $<$

M1

$$|x| < \frac{3}{5} \text{ or } -\frac{3}{5} < x < \frac{3}{5}$$

CAO

A1

2

[12]

Q7.

(a) $1 = A(1 - x)^2 + B(1 - x)(3 - 2x) + C(3 - 2x)$
 Attempt to clear fractions

M1

$$\left. \begin{array}{lll} x = 1 & x = \frac{3}{2} & x = 0 \\ C = 1 & 1 = A\left(-\frac{1}{2}\right)^2 & 1 = A + 3B + 3C \end{array} \right\}$$

Use any two (or three) values of x to set up
 two (or three) equations

m1

$$A = 4 \quad B = -2 \quad C = 1$$

Two values correct

A1

All values correct

A1

Alternative

$$1 = A(1 - x)^2 + B(1 - x)(3 - 2x) + C(3 - 2x)$$

(M1)

$$1 = A + 3B + 3C$$

$$0 = -2A - 5B - 2C$$

Set up three simultaneous equations

(m1)

$$0 = A + 2B$$

Two values correct

(A1)

$$A = 4 \quad B = -2 \quad C = 1$$

All values correct

(A1)

4

(b)
$$\int \frac{1}{2\sqrt{y}} dy = \int \frac{4}{3 - 2x} - \frac{2}{1 - x} + \frac{1}{(1 - x)^2} dx$$

*Separate using partial fractions; correct notation;
condone missing integral signs but dy and dx
must be in correct place.*

ft on their A, B, C and on each integral.

B1ft

$$\int \frac{1}{2\sqrt{y}} dy = \sqrt{y} =$$

OE
$$\int \frac{k}{\sqrt{y}} dy = 2k\sqrt{y}$$
 is B1

B1

$$-2 \ln(3 - 2x)$$

Condone missing brackets on one ln integral.

B1ft

$$+ 2 \ln (1 - x)$$

B1ft

$$+ \frac{1}{1-x} (+ C)$$

Condone omission of +C

B1ft

$$x = 0 \quad y = 0 \Rightarrow 0 = -2 \ln 3 + 0 + 1 + C$$

Use (0,0) to find C. Must get to C =

M1

$$C = 2 \ln 3 - 1$$

Correct C found from correct equation.

C must be exact, in any form but not decimal.

A1

$$\sqrt{y} = 2 \ln \left(\frac{3-3x}{3-2x} \right) + \frac{1}{1-x} - 1$$

Correct use of rules of logs to progress towards requested form of answer. C must be of the form $r \ln s + t$

m1

$$y^{\frac{1}{2}} = 2 \ln \left(\frac{3-3x}{3-2x} \right) + \frac{x}{1-x}$$

OE 2025 Exam Papers Practice. All rights reserved.
CSO condone B0 for separation

A1

9

[13]

Q8.

(a) $10x^2 + 8 = 2(x + 1)(5x - 1) +$

A and B terms correct

M1

$$A(5x - 1) + B(x + 1)$$

A1

$$x = -1 \quad x = \frac{1}{5}$$

$$A = -3 \quad B = 7$$

Use two values of x to find A and B , or
set up and solve

$$8 + 5A + B = 0$$

$$-2 - A + B = 8$$

SC1 NWS A & B correct $\frac{4}{4}$

SC2 NWS A or B correct $\frac{1}{4}$

m1A1

4

$$(b) \quad \int \frac{10x^2 + 8}{(x+1)(5x-1)} dx = \int 2 - \frac{3}{x+1} + \frac{7}{5x-1} dx$$

Use the partial fractions

$$= 2x + C$$

M1

B1

$a \ln(x+1) + b \ln(5x-1)$
condone missing brackets

M1

$$-3 \ln(x+1) + \frac{7}{5} \ln(5x-1)$$

F on A and B Exam Papers Practice. All rights reserved.

A1F

4

[8]

Q9.

$$(a) \quad (i) \quad 7x - 3 = A(3x - 2) + B(x + 1)$$

M1

$$x = -1 \quad x = \frac{2}{3}$$

Substitute two values of x and solve for A and B

m1

$$A = 2 \quad B = 1$$

$$\text{Or solve } \left. \begin{array}{l} 7 = 3A + B \\ -3 = -2A + B \end{array} \right\} \text{ condone one error}$$

A1

Alternative:

By cover up rule

$$x = -1 \quad A = \frac{-7-3}{-5}$$

$$x = \frac{2}{3} \quad B = \frac{\frac{14}{3}-3}{\frac{5}{3}}$$

$$x = -1 \text{ and } x = \frac{2}{3}$$

and attempt to find A and B

(M1)

$$A = 2 \quad B = 1$$

SC NMS A and B both correct 3/3

One of A or B correct 1/3

(A1, A1)

3

$$(ii) \int \frac{7x-3}{(x+1)(3x-2)} dx =$$

$$p \ln(x+1) + q \ln(3x-2)$$

Condone missing brackets

M1

$$= 2 \ln(x+1) + \frac{1}{3} \ln(3x-2) (+c)$$

F on A and B; constant not required

A1F

2

$$(b) \frac{6x^2 + x + 2}{2x^2 - x + 1} = \frac{6x^2 - 3x + 3 + 4x - 1}{2x^2 - x + 1}$$

M1

$$= 3 + \frac{4x-1}{2x^2-x+1}$$

$$P = 3$$

B1

$$Q = 4 \text{ and } R = -1$$

A1

Alternative:

$$\begin{array}{r}
 3 \\
 2x^2 - x + 1 \overline{) 6x^2 + x + 2} \\
 \underline{6x^2 - 3x + 3} \\
 4x - 1
 \end{array}$$

Complete division, with $ax + b$ remainder

(M1)

$$P = 3 \text{ stated}$$

(B1)

$Q = 4$ and $R = -1$ stated or written as expression

(A1)

or

$$\begin{aligned}
 6x^2 + x + 2 &= P(2x^2 - x + 1) + Qx + R \\
 &= 2Px^2 + (Q - P)x + P + R
 \end{aligned}$$

Multiply across and equate coefficients or use numerical values of x

(M1)

$$P = 3$$

$$P = 3 \text{ stated}$$

(B1)

$$Q - P = 1$$

$$P + R = 2$$

$$Q = 4 \text{ and } R = -1$$

$Q = 4$ and $R = -1$ stated or written as expression

(A1)

3

[8]

Q10.

$$(a) \quad (i) \quad \frac{2x+7}{x+2} = 2 + \frac{3}{x+2}$$

B1B1

2

(ii) $\int \frac{2x+7}{x+2} = 3 \ln(x+2) + 2x + C$

Either term correct

B1F

*Both correct; constant required; condone missing bracket
ft on A, B*

B1F

2

(b) (i) $28 + 4x^2$
 $P(5-x)^2 + Q(1+3x)(5-x) + R(1+3x)$

M1

$x = 5$ $x = -\frac{1}{3}$

Two values of x used to find R and P.

m1

$R = 8$ $P = 1$

SC R = 8, P = 1 NMS can score B1, B1

A1

$x = 0 \Rightarrow 28 = 25P + 5Q + R$

Third value of x used to find Q

©2025 Exam Papers Practice. All rights reserved. m1

$Q = -1$

A1

Alternative

$28 + 4x^2 =$

$P(5-x)^2 + Q(1-3x)(5-x) + R(1+3x)$

(M1)

$= (25P + 5Q + R) + (-10P + 14Q + 3R)x + (P - 3Q)x^2$

Collect terms and form equations

(m1)

$P - 3Q = 4$

$$14Q + 3R - 10P = 0$$

Correct equations

(A1)

$$25P + 5Q + R = 28$$

$$P = 1 \quad Q = -1 \quad R = 8$$

Solve for P Q and R

(m1)(A1)

5

(ii) $\int \frac{1}{1+3x} - \frac{1}{5-x} + \frac{8}{(5-x)^2} dx$

Use partial fractions

M1

$$= \frac{1}{3} \ln(1+3x) + \ln(5-x) + \frac{8}{5-x} + (C)$$

$$a \ln(1+3x) + b \ln(5-x)$$

m1

OE; both ln integrals correct; needs ()

A1F

*Other term correct
ft on their P, Q, R*

A1F

4

*SC: If no P, Q, R found in (b)(i), can gain
method marks by inserting other values or
retaining the letters (max 2/4)*

[13]

Q11.

$$3x-1 \overline{) 3x^3 + 8x^2 - 3x - 5}$$

$$\begin{array}{r} 3x^3 \quad -x^2 \\ \underline{9x^2 - 3x} \\ 9x^2 - 3x \end{array}$$

Division with x^2 and an x term seen; $x^2 + px$

M1

$$a = 1 \quad b = 3 \quad \text{or} \quad x^2 + 3x + \frac{c}{3x-1}$$

Explicit or in expression

A1

$$c = -5$$

$$\text{Condone} + \frac{-5}{3x-1}$$

B1

Alternative

$$\frac{(3x-1)(x^2 + px)}{3x-1} - \frac{5}{3x-1}$$

Split fraction and attempt factors

(M1)

$$x^2 + 3x - \frac{5}{3x-1}$$

$$a = 1 \quad b = 3$$

(A1)

$$c = -5$$

(B1)

Alternative

$$f(x) = 3ax^3 + (3b - a)x^2 - bx + c$$

Multiply by (3x - 1) and attempt to collect terms

(M1)

$$a = 1 \quad b = 3$$

(A1)

$$c = -5$$

(B1)

Alternative

$$f(x) = (ax^2 + bx)(3x - 1) + c$$

Multiply by (3x - 1) and attempt to find a, b, c: substitute 3 values of x and form 3 simultaneous equations, and attempt to solve; or substitute 3 values of x into given equation

(M1)

$$x = 0 \Rightarrow c = -5$$

(B1)

$$x = 1 \Rightarrow 2a + 2b + c = 3$$

$$x = 2 \Rightarrow 20a + 10b + c = 45$$

$$a = 1 \quad b = 3$$

(A1)

[3]

Q12.

(a) $(1-x)^2 = 1 + (-1)(-x) + \frac{1}{2}(-1, -2)(-x)^2$

$$1 \pm x + kx^2$$

M1

$$= 1 + x + x^2$$

Fully simplified

A1

2

(b) (i) $3x - 1 = A(2 - 3x) + B(1 - x)$

M1

$$x = 1 \quad x = \frac{2}{3}$$

*Use 2 values of x or equate coefficients
and solve $-3A - B = 3$ $2A + B = -1$
condone coefficient errors*

m1

$$A = -2 \quad B = 3$$

Both values

NMS 3/3 if both correct, 1/3 if one correct

A1

3

(ii) $\left(\frac{3x-1}{(1-x)(2-3x)} = \frac{-2}{1-x} + \frac{3}{2-3x} \right)$

$$\frac{-2}{1-x} = -2 - 2x - 2x^2$$

F on $(1 - x)^{-1}$ and A

B1F

$$\frac{1}{2 - 3x} = \frac{1}{2} \left(1 - \frac{3}{2}x \right)^{-1}$$

B1

$$= (p)(1 + kx + (kx)^2)$$

$$p, k = \text{candidate's } \frac{1}{2}, \frac{3}{2}, k \neq \pm 1$$

M1

$$= (p) \left(1 + \frac{3}{2}x + \frac{9}{4}x^2 \right)$$

Use (a) or start binomial again; condone missing brackets, and one sign error

A1

$$\frac{3x - 1}{(1 - x)(2 - 3x)} = -2(1 - x)^{-1} + 3(2 - 3x)^{-1}$$

Valid combination of both expansions

M1

$$= -\frac{1}{2} + \frac{1}{4}x + \frac{11}{8}x^2$$

CSO

A1

Alternative

$$(2 - 3x)^{-1} = \frac{1}{2} \left(1 - \frac{3}{2}x \right)^{-1}$$

$$k = \text{candidate's } \frac{3}{2} \quad k \neq \pm 1$$

(B1)

$$(1 - kx)^{-1} = 1 + kx + (kx)^2$$

Use (a) or start binomial again; condone missing brackets and one error

(M1)

$$= 1 + \frac{3}{2}x + \frac{9}{4}x^2$$

(A1)

$$\frac{3x-1}{(1-x)(2-3x)} = (3x-1)(1-x)^{-1}(2-3x)^{-1}$$

(3x - 1) × both expansions

(M1)

$$\frac{3x-1}{(1-x)(2-3x)} = -\frac{1}{2} + \frac{1}{4}x + \frac{11}{8}x^2$$

*Multiply out; collect terms to form
a + bx + cx²*

(m1)

CSO

(A1)

Alternative for (2 - 3x)⁻¹

Using (a + bx)ⁿ

$$2^{-1} + (-1)(2)^{-2}(-3x) + \frac{(-1)(-2)(2)^{-3}(-3x)^2}{2}$$

Condone missing brackets, and 1 error

(M1)

$$= \frac{1}{2} + \frac{3}{4}x + \frac{9}{8}x^2$$

First two terms

(A1)

x² term

(A1)

6

(c) $-2 < 3x < 2$

PI, or any equivalent form

M1

$$\Rightarrow -\frac{2}{3} < x < \frac{2}{3}$$

Condone $\leq$; accept $x \geq \frac{2}{3}$ or $x \geq -\frac{2}{3}$

CSO; allow $|\pm x| \leq \frac{2}{3}$, or

$$x < \frac{2}{3} \text{ and } x > -\frac{2}{3}$$

A1

2

[13]

Q13.

(a) $3 = k(3 + x + 3 - x)$

OE $\frac{A}{3-x} + \frac{B}{3+x} \Rightarrow 6A = 3 \quad 6B = 3$

M1

$$k = \frac{1}{2}$$

or eg put $x = 0, \frac{3}{9} = k\left(\frac{1}{3} + \frac{1}{3}\right) \Rightarrow k = \frac{1}{2}$

A1

2

(b) $\int_1^2 \frac{3}{9-x^2} dx = -\frac{1}{2} \ln(3-x) + \frac{1}{2} \ln(3+x)$

$a \ln(3 \pm x)$

M1

ft on k

A1F

$$= \frac{1}{2} ((\ln 5 - \ln 1) - (\ln 4 - \ln 2)) = \frac{1}{2} \ln\left(\frac{5}{2}\right)$$

accept $\ln\left(\frac{10}{4}\right)$

ft only for sign error in integral: $\frac{1}{2} \ln\left(\frac{5}{8}\right)$

A1F

3

Q14.

(a) (i) $f\left(\frac{1}{2}\right) = 2 \times \left(\frac{1}{2}\right)^3 + 3 \times \left(\frac{1}{2}\right)^2 - 18\left(\frac{1}{2}\right) + 8$

use of $\pm \frac{1}{2}$ substituted in $f(x)$

M1

$$= \frac{1}{4} + \frac{3}{4} - 9 + 8 = 0 \Rightarrow \text{factor}$$

arithmetic seen and conclusion –

minimum seen: $2 \times \frac{1}{8} + 3 \times \frac{1}{4} - 18 \times \frac{1}{2} + 8 = 0$

A1

2

(ii) $f(x) = (2x - 1)(x^2 + 2x - 8)$

or $p = 2, q = -8$

B1B1

2

(iii) $\frac{4x(x+4)}{(2x-1)(x+4)(x-2)}$

numerator correct; attempt to factorise
denominator (algebraic fraction not required)

M1

$$= \frac{4x}{(2x-1)(x-2)}$$

CAO

A1

2

(b) $2x^2 = A(x+5)(x-3) + B + Cx$

any equivalent method
using PFs (see alternative method)

M1

$A = 2$

B1

$$2A + C = 0 \quad -15A + B = 0$$

equate coefficients or use 2 values of x to find B and C

M1

$$C = -4 \quad B = 30$$

both B and C correct

A1

ALTERNATIVE METHOD 1

$$\begin{array}{r}
 x^2 + 2x - 15 \overline{) 2x^2} \\
 \underline{2x^2 + 4x - 30} \\
 -4x + 30
 \end{array}$$

complete division

(M1)

$$A = 2$$

(B1)

$$B = 30$$

(A1)

$$C = -4$$

(A1)

ALTERNATIVE METHOD 2

$$\frac{2x^2}{(x+5)(x-3)} = A + \frac{D}{x+5} + \frac{E}{x-3}$$

$$2x^2 = A(x+5)(x-3) + D(x-3) + E(x+5)$$

$$x = 3 \quad 18 = 8E \quad E = \frac{9}{4}$$

$$x = -5 \quad 50 = -8D \quad D = -\frac{25}{4}$$

find D and E

(M1)

$$x = 0, 0 = -15A + \left(-\frac{25}{4}\right)(-3) + \left(\frac{9}{4}\right)(5)$$

$$A = 2$$

(B1)

$$\begin{aligned}
 \frac{D}{x+5} + \frac{E}{x-3} &= \frac{-25}{4(x+5)} + \frac{9}{4(x-3)} \\
 &= \frac{-25(x-3) + 9(x+5)}{4(x+5)(x-3)} \\
 &= \frac{120 - 16x}{4(x+5)(x-3)}
 \end{aligned}$$

recombine to required form

(M1)

$$= \frac{30 - 4x}{(x+5)(x-3)}$$

CAO

(A1)

4

[10]

Q15.

(a)
$$\frac{2}{(x^2 - 1)} = \frac{A}{x-1} + \frac{B}{x+1}$$

$$2 = A(x+1) + B(x-1)$$

M1

$$x = 1 \qquad x = -1$$

*use two values of x or equate coefficients and
solve $A + B = 0$ and $A - B = 2$*

m1

$$A = 1 \qquad B = -1$$

both A and B

A1

3

(b)
$$\int \frac{2}{x^2 - 1} dx = p \ln(x-1) + q \ln(x+1)$$

In integrals

M1

$$= \ln(x-1) - \ln(x+1)$$

F on A and B

condone missing brackets

A1F

2

(c) $\int \frac{dy}{y} = \int \frac{2}{3(x^2 - 1)} dx$

separate and attempt to integrate on one side

M1

$$\ln y = \frac{1}{3} (\ln(x-1) - \ln(x+1)) + C$$

left hand side

A1

F from part (b) on right hand side

A1F

(3, 1) $\ln 1 = \frac{1}{3} (\ln 2 - \ln 4) + C$

use (3, 1) to attempt to find a constant

m1

$$3 \ln y = \ln(x-1) - \ln(x+1) - (\ln 2 - \ln 4)$$

$$3 \ln y = \ln \left(\frac{x-1}{x+1} \right) + \ln 2$$

$$\ln y^3 = \ln \left(\frac{2(x-1)}{x+1} \right)$$

$$y^3 = \frac{2(x-1)}{x+1}$$

CSO AG

A1

5

[10]

Q16.

(a) (i) $\frac{3x-5}{x-3} = 3 + \frac{4}{x-3}$

*By division: B1 for 3, B1 for $\frac{4}{x-3}$ or $B = 4$
By partial fractions: M1 multiply by $x-3$
and using 2 values of x , A1 both correct*

B1, B1

2

(ii) $\int 3 + \frac{4}{x-3} dx = 3x + 4 \ln(x-3) (+c)$

M1 $\int 3 + \frac{4}{x-3} dx$ and attempt at integrals
ft on A and B; condone omission of
brackets around $x - 3$

M1A1F

2

Alternative: By substitution $u = x - 3$

$$\int \frac{3x-5}{x-3} dx = \int \frac{3u+4}{u} du$$

Integral in terms of u

(M1)

$$= 3(x-3) + 4 \ln(x-3)$$

Correct, in x

(A1)

(b) (i) $6x - 5 = P(2x - 5) + Q(2x + 5)$

Clear evidence of use of cover-up rule M2

M1

EXAM PAPERS PRACTICE

$$x = \frac{5}{2} \quad x = -\frac{5}{2}$$

©2025 Exam Papers Practice. All rights reserved.

m1

$$10 = 10Q - 20 = -10P$$

$$Q = 1 \quad P = 2$$

A1

3

(ii) $\int \frac{2}{2x+5} + \frac{1}{2x-5} dx$

Attempt at ln integral
($a \ln(2x + 5) + b \ln(2x - 5)$)

M1

$$\ln(2x+5) + \frac{1}{2} \ln(2x-5) (+c)$$

M1

ft on P and Q; must have brackets

A1F

3

[10]

Q17.

(a) (i) $(1+x)^{-1} = 1 + (-1)x + px^2 + qx^3$
 $p \neq 0, q \neq 0$

M1

$$= 1 - x + x^2 - x^3$$

$$SC \ 1/2 \text{ for } = 1 - x + px^2$$

A1

2

(ii) $(1+3x)^{-1} = 1 - 3x + (3x)^2 - (3x)^3$
x replaced by 3x in candidate's (a)(i); condone missing brackets

M1

$$= 1 - 3x + 9x^2 - 27x^3$$

EXAM PAPERS PRACTICE

CAO SC x^3 -term: $\frac{1-3x+\frac{3}{9}x^2}{1/2}$

©2025 Exam Papers Practice. All rights reserved. A1

2

Alt (starting again)

$$(1+3x)^{-1} = 1 - (3x) + \frac{(-1)(-2)(3x)^2}{2!} + \frac{(-1)(-2)(-3)(3x)^3}{3!}$$

condone missing brackets
accept 2 for 2!, 3.2 for 3!

(M1)

$$= 1 - 3x + 9x^2 - 27x^3$$

CAO

(A1)

(2)

(b)
$$\frac{1+4x}{(1+x)(1+3x)} = \frac{A}{1+x} + \frac{B}{1+3x}$$

*correct partial fractions form, and
multiplication by denominator*

M1

$$1 + 4x = A(1 + 3x) + B(1 + x)$$

$$x = -1, \quad x = -\frac{1}{3}$$

Use (any) two values of x to find A and B

m1

$$A = \frac{3}{2}, \quad B = -\frac{1}{2}$$

A and B both correct

A1

3

Alt:

$$\frac{1+4x}{(1+x)(1+3x)} = \frac{A}{1+x} + \frac{B}{1+3x}$$

*correct partial fractions form, and
multiplication by denominator*

(M1)

$$1 + 4x = A(1 + 3x) + B(1 + x)$$

$$A + B = 1, \quad 3A + B = 4$$

Set up and solve

(m1)

$$A = \frac{3}{2}, \quad B = -\frac{1}{2}$$

A and B both correct

(A1)

(3)

(c) (i)
$$\frac{1+4x}{(1+x)(1+3x)} = \frac{3}{2(1+x)} - \frac{1}{2(1+3x)}$$

M1

$$= \frac{3}{2}(1 - x + x^2 - x^3) - \frac{1}{2}(1 - 3x + 9x^2 - 27x^3)$$

multiply candidate's expansions by A and B, and expand and simplify

m1

$$= 1 - 3x^2 + 12x^3$$

CAO

SC A and B interchanged, treat as miscopy. $(1 - 4x + 13x^2 - 40x^3)$

A1

3

Alt

$$\frac{1+4x}{(1+x)(1+3x)} = (1+4x)(1+x)^{-1}(1+3x)^{-1}$$

$$= (1+4x)(1-x+x^2-x^3)(1-3x+9x^2-27x^3)$$

write as product, using expansions condone missing brackets on $(1+4x)$ only

(M1)

$$= 1 - 4x + 13x^2 - 40x^3 + 4x - 16x^2 + 52x^3$$

attempt to multiply the three expansions up to terms in x^3

(m1)

$$= 1 - 3x^2 + 12x^3$$

CAO

(A1)

(3)

(ii) $|x| < 1$ and $|3x| < 1$

OE and nothing else incorrect

M1

$$|x| < \frac{1}{3} \quad (0.33)$$

OE Condone $\leq$

A1

2

[12]

Q18.

(a) (i) $(1 - x)^{-1} = 1 + (-1)(-x) + \frac{(-1)(-2)}{2} (-x)^2$
First two terms + kx^2

M1

$$= 1 + x + x^2$$

A1

2

(ii) $\frac{1}{(3 - 2x)} = \frac{1}{3} \left(1 - \frac{2}{3}x \right)^{-1}$

Or directly substitute into formula;

B1

$$\approx * \left(1 + \frac{2}{3}x + \left(\frac{2}{3}x \right)^2 \right)$$

M1 power of 3

M1 other coefficients (allow one error)

A1 CAO

M1

$$\approx \frac{1}{3} + \frac{2}{9}x + \frac{4}{27}x^2$$

AG convincingly obtained

A1

3

(b) $(1 - x)^{-2} = 1 + (-2)(-x) + \frac{(-2)(-3)(-x)^2}{2}$

M1

$$= 1 + 2x + 3x^2$$

A1

2

(c) $2x^2 - 3 = A(1 - x)^2 + B(3 - 2x)(1 - x) + C(3 - 2x)$

M1

$$x = 1 \quad -1 = C \times 1 \quad x = \frac{3}{2} \quad \frac{3}{2} = A \times \frac{1}{4}$$

Or by equating coefficients

M1 same

A1 collect terms

M1 equate coefficients

A1 2 correct

A1 3 correct

M1

$$C = -1 \quad A = 6$$

Follow on A and C

A1

$$x = 0 \quad (-3 = 6 + 3B - 3)$$

or other value $\Rightarrow$ equation in A, B, C

m1

$$B = -2$$

A1

5

(d)
$$\frac{6}{3-2x} - \frac{2}{1-x} - \frac{1}{(1-x)^2}$$

$$\approx \frac{6}{3} \left(1 + \frac{2}{3}x + \frac{4}{9}x^2 \right) - 2(1 + x + x^2)$$

Follow on A B C and expansions

M1A1F

$$-(1 + 2x + 3x^2) \approx -1 - \frac{8}{3}x - \frac{37}{9}x^2$$

CAO

A1

3

[15]

Q19.

(a) $9x^2 - 6x + 5 = 3(3x - 1)(x - 1) + A(x - 1) + B(3x - 1)$

$$\text{Or } 3 + \frac{6x+2}{(3x-1)(x-1)}$$

B1

$$x = 1 \quad x = \frac{1}{3}$$

Substitute $x = 1$ or $x = \frac{1}{3}$

M1

$$B = 4 \quad A = -6$$

Or equivalent method (equating coefficients, simultaneous equations)

A1A1

4

(b) $\int = \int 3 - \frac{6}{3x-1} + \frac{4}{x-1} dx$

Attempt to use partial fractions

M1

$$= 3x \dots$$

B1

$$-2 \ln(3x-1) + 4 \ln(x-1) (+c)$$

$p \ln(3x-1) + q \ln(x-1)$
Condone missing brackets

M1

Follow through on A and B; brackets needed.

A1F

4

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

[8]

Q20.

(a) $3x - 5 = A(2x - 1) + B(x + 3)$

M1

$$x = -3 \quad A = 2, \quad x = \frac{1}{2} \quad B = -1$$

m1: sub in 2 values or set up simultaneous equations

m1A1

3

(b) $\int \left(\frac{2}{x+3} - \frac{1}{2x-1} \right) dx$

M1

$$= 2 \ln (x+3) - \frac{1}{2} \ln (2x-1) + c$$

Use of lns for either integral

m1

ft A and B

A1ft

3

[6]



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.